

DSA ASSIGNMENT QUESTIONS

1. Write a pseudocode for storing and addition of these 2 polynomials and return the final polynomial using linked list. 1st polynomial: $2x^8 - 5x^7 - 3x^2 + 4$ and 2nd polynomial: $x^7 + 5x^2 + x + 3$.
2. Write a pseudocode to traverse a circular linked list (a simple linked list with last node pointing to the first node).
3. Reverse linked list with space complexity of $O(1)$.
4. Implement 3 stacks in a single array.
5. Implement merge sort and show the merging to sort the following array in ascending order:
351, 285, 310, 254, 179, 861, 423, 652, 450, 520.
6. Create a recursion tree for tower of Hanoi for $n = 4$.
7. We used a command of "goto" in while loop, while implementing in-order traversal of a binary tree. Avoid using this "goto" and modify this iterative algorithm.
8. If we wanna remove duplicate elements in an array, how? If we want space complexity to be $O(1)$, also calculate, what will be the time complexity? **Array is: 14, 10, 17, 12, 10, 11, 20, 12, 18, 25, 20, 8, 22, 11, 23.**
9. Solve Q6 by trying sorting the array and using if condition.
10. Let's say, $O(1)$ is not there, then how to solve Q6.
11. Compare total number of comparisons done in our solution of Q6 and if BST were implemented to solve the question.
12. Write function for extractmin, which is written in pseudocode of Huffman code.
13. Also implement the above said extractmin function in heap.
14. Given a 5-way B-tree created by the data: **3, 7, 9, 23, 45, 1, 5, 4, 14, 25, 24, 13, 11, 8, 9, 4, 31, 55, 56.** Add these further keys: **2, 6, 12** and Delete these keys: **4, 5, 7, 3, 14.**
15. Implement **UNION** function and then apply in **KRUSKAL ALGORITHM**.